

BDF - Customized Product Documentation

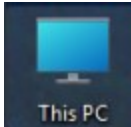
BDF EyeQ Application Documentation

This documentation explains all the pages and their functionalities for the BDF EyeQ application, designed to support your bakery automation processes.

Running the Application

On the Current PC

To run the BDF EyeQ application on the current computer, double-click the desktop shortcut named `ABIUnifiedApp`.



This PC



ABIUnified...



Recycle Bin



Docker
Desktop



Learn about
this image



SLTools



AnyDesk



Microsoft
Edge



Notepad++



Pylon Viewer



RustDesk



UaExpert

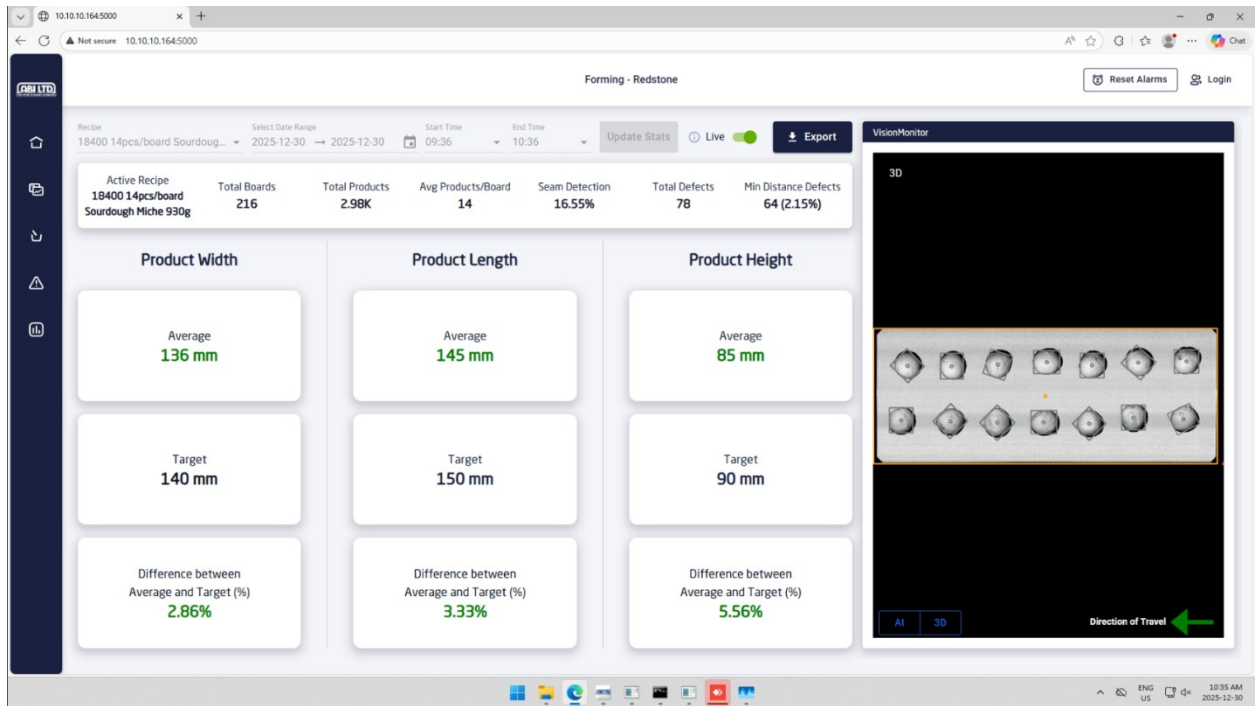


Inference_I...

Accessing from Another PC on the Local Area Network (LAN)

The BDF EyeQ application typically runs on a primary control computer (the "host"), but it can be accessed from other computers within the same local area network (LAN) using a standard web browser. A LAN is simply a network of computers connected together in the same office or production area.

To access the application from another PC, open your web browser and enter the host machine's network IP address and port number. If your system has multiple production points, specific web addresses for each point will be provided by your BDF system administrator. We recommend bookmarking these links for easy access.



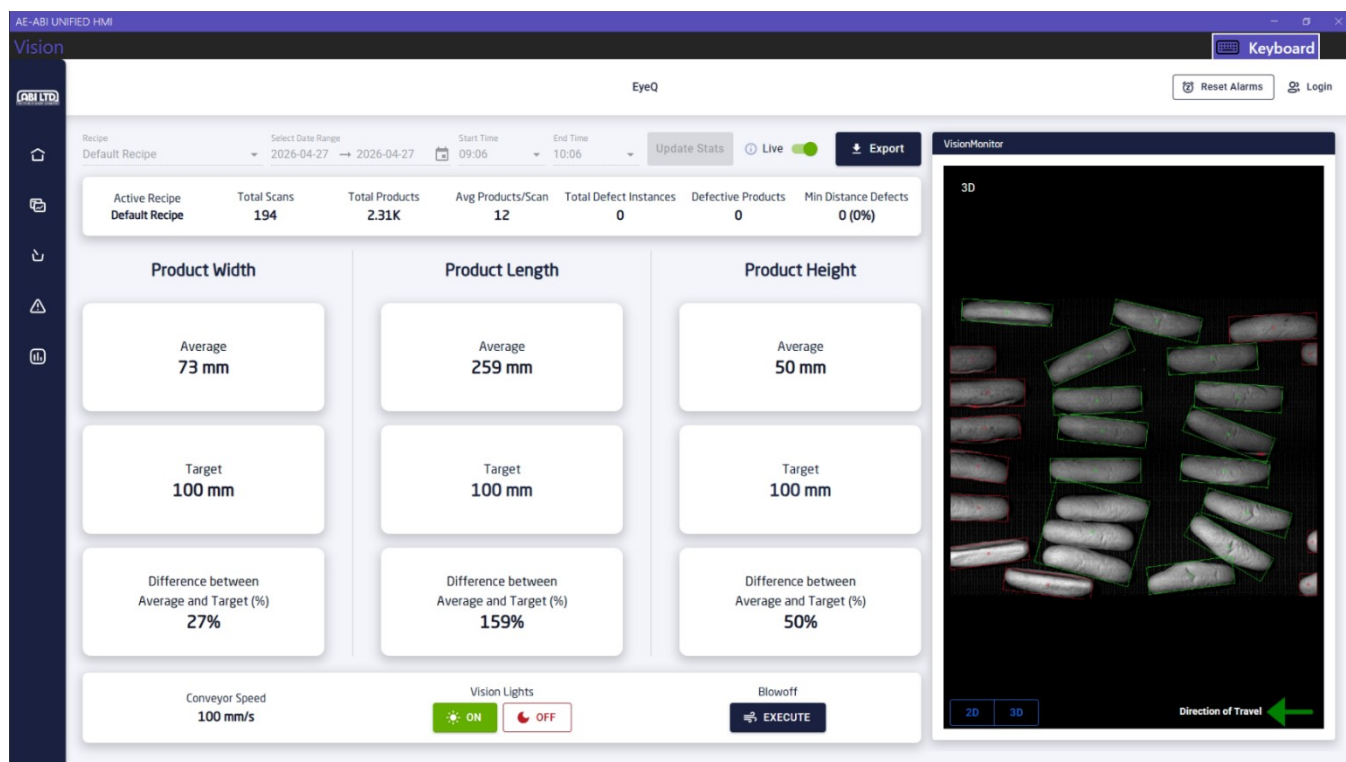
Application Pages

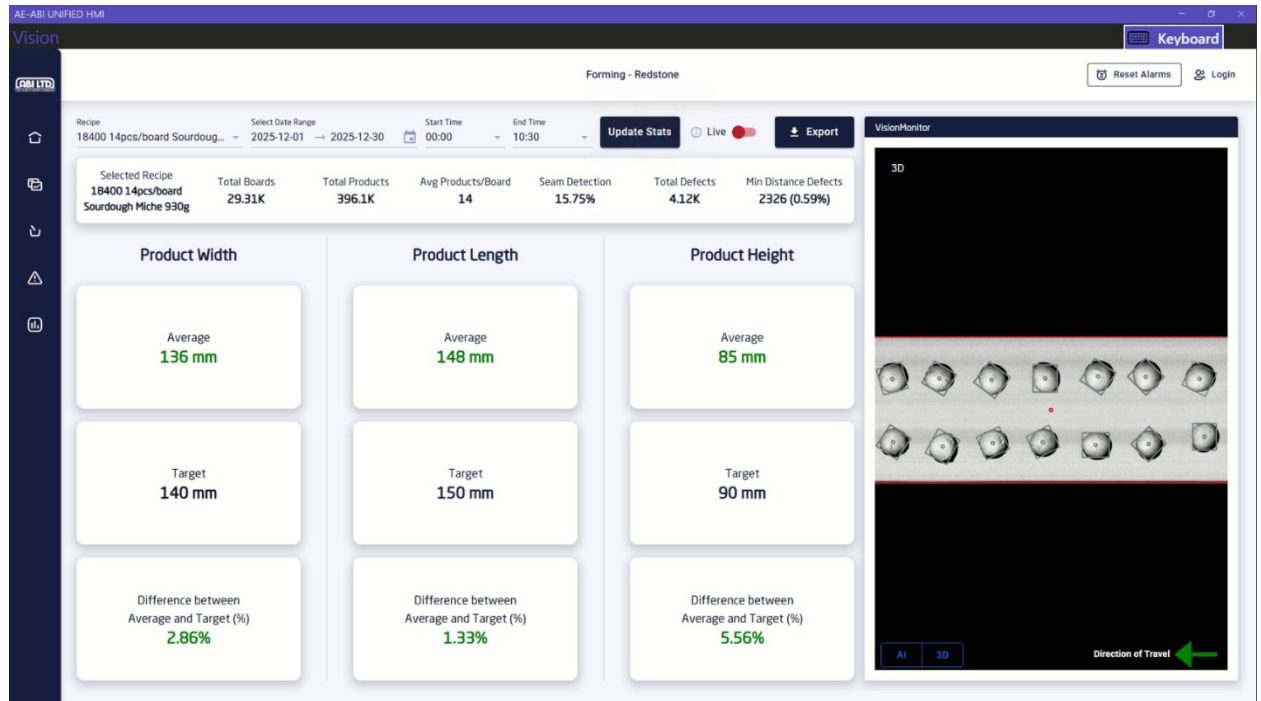
Home

The Home page provides a clear overview of your production line's current status and historical data.

- **Active Recipe:** Clearly shows the product recipe currently being used by the system.
- **Performance Statistics (Last Hour):** Displays key metrics for the past hour, including:
 - The number of scans or production boards processed.
 - Total number of products inspected.
 - Average products inspected per scan or board.
 - Total number of defects detected.
 - The number of individual products found to be defective.

- **Product Measurements:** Presents the average, target, and percentage difference for critical product measurements such as width, length, and height. These statistics are shown in green if they are within acceptable limits (threshold values) and in red if they are outside these limits.
- **Vision Monitor:** On the right side of the page, a vision monitor displays a 3D image of the most recently inspected product scan or production board. A 2D or AI (Artificial Intelligence) tab may also show a live view of the conveyor with products, processed images, or specific defect detections (e.g., seam detections). It also indicates the direction of movement for the conveyor and products. If configured, it can highlight areas where the distance between products falls below a set minimum.
- **Historical Data:** To view data from more than an hour ago, toggle the "Live" button off. You can then select a specific Recipe, Date Range, Start Time, and End Time, and click "Update Stats" to load the historical information. These fields are initially disabled when the "Live" view is active.
- **Export Data:** Use the "Export" button to save all product data for your selected recipe and timeframe as a CSV (comma separated values) file, which can be opened in spreadsheet programs.

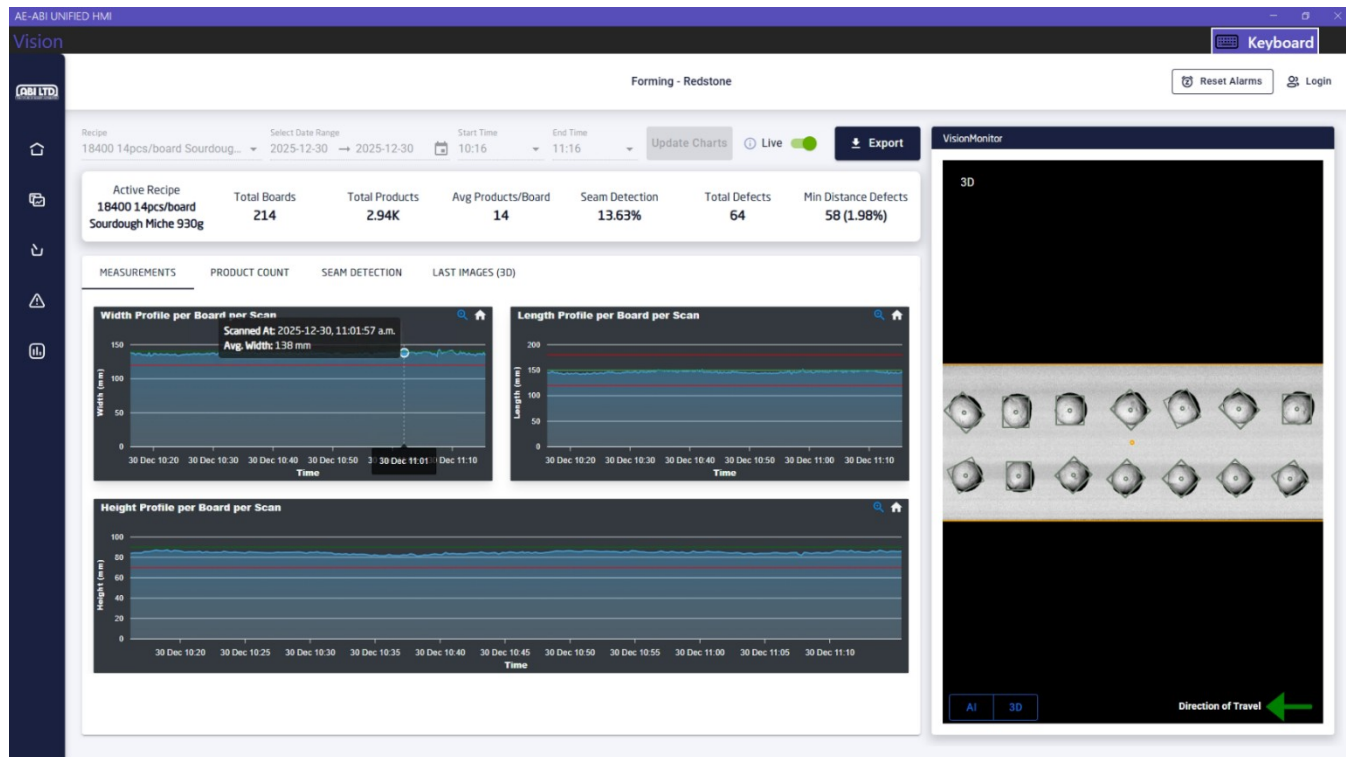




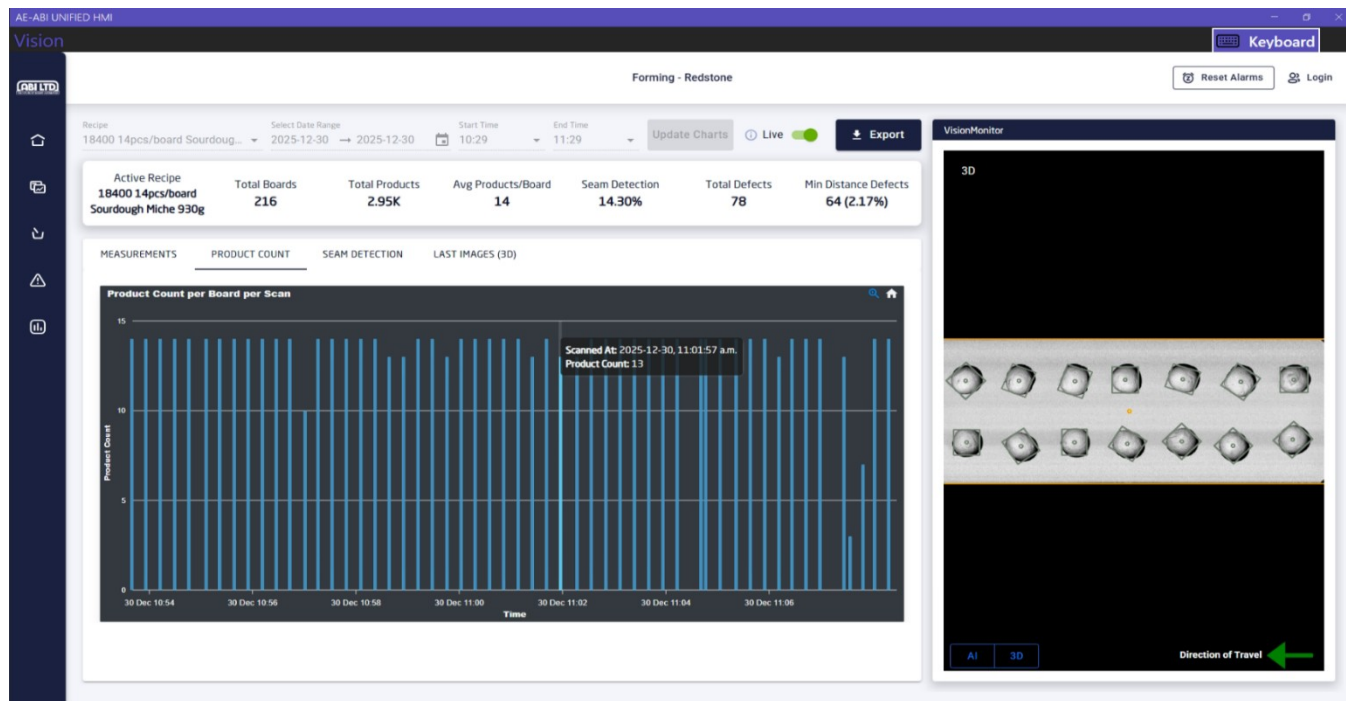
Trends

The Trends page offers detailed charts and graphs for various production parameters over time. The filtering and live toggling functions work similarly to the Home page.

- **Measurements:** This tab shows the average width, length, and height of products per scan or board over time. You can zoom in on a specific period by selecting and dragging a portion of the chart. The home icon can be pressed to reset the chart to its default view.



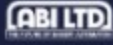
- **Product Count:** This tab displays the number of products processed per scan or board over time.



- **Product Appearance Analysis:** This tab provides insights into the visual characteristics of your bakery products.

- **L-Channel Analysis (if configured):** Shows the average L^* values (a measure of lightness or darkness) for products over time, calculated using 2D cameras.

Vision



Recipe

Default Recipe

Select Date Range

2026-04-27 → 2026-04-27

Start Time

09:15

Active Recipe
Default Recipe

Total Scans
285

Total Products
3.48K

Avg Products/Scan
12

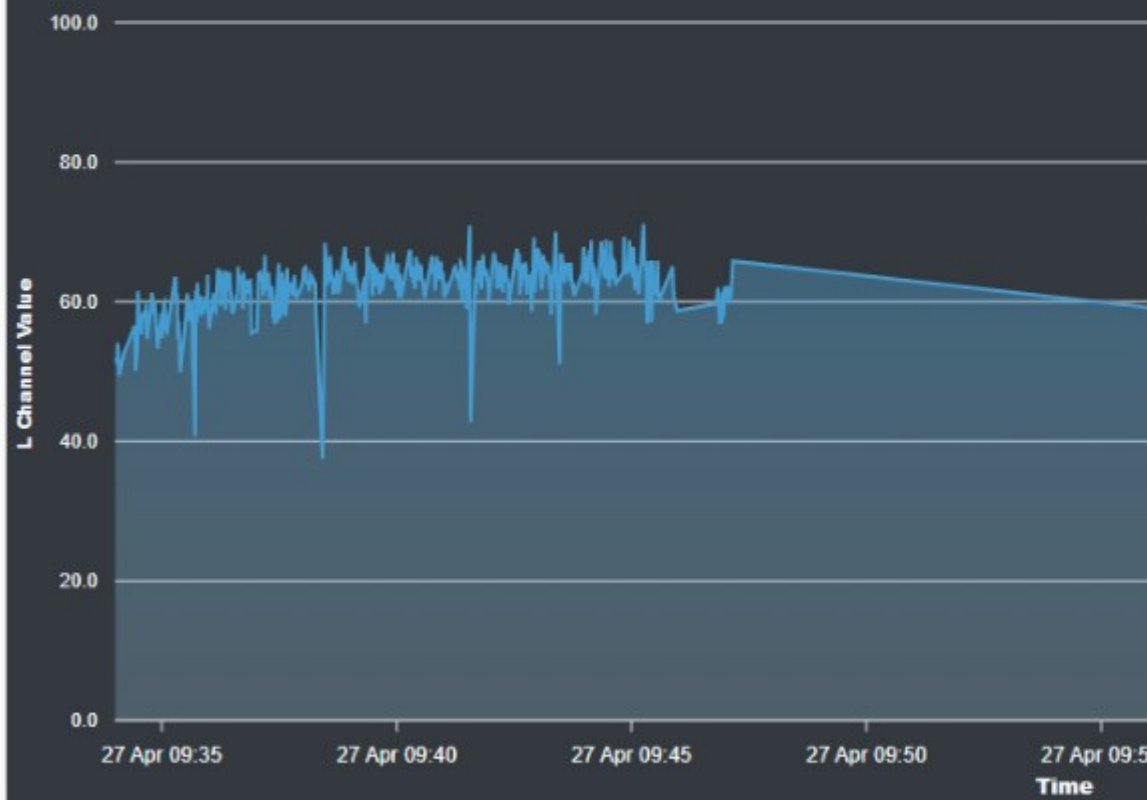
MEASUREMENTS

L CHANNEL ANALYSIS

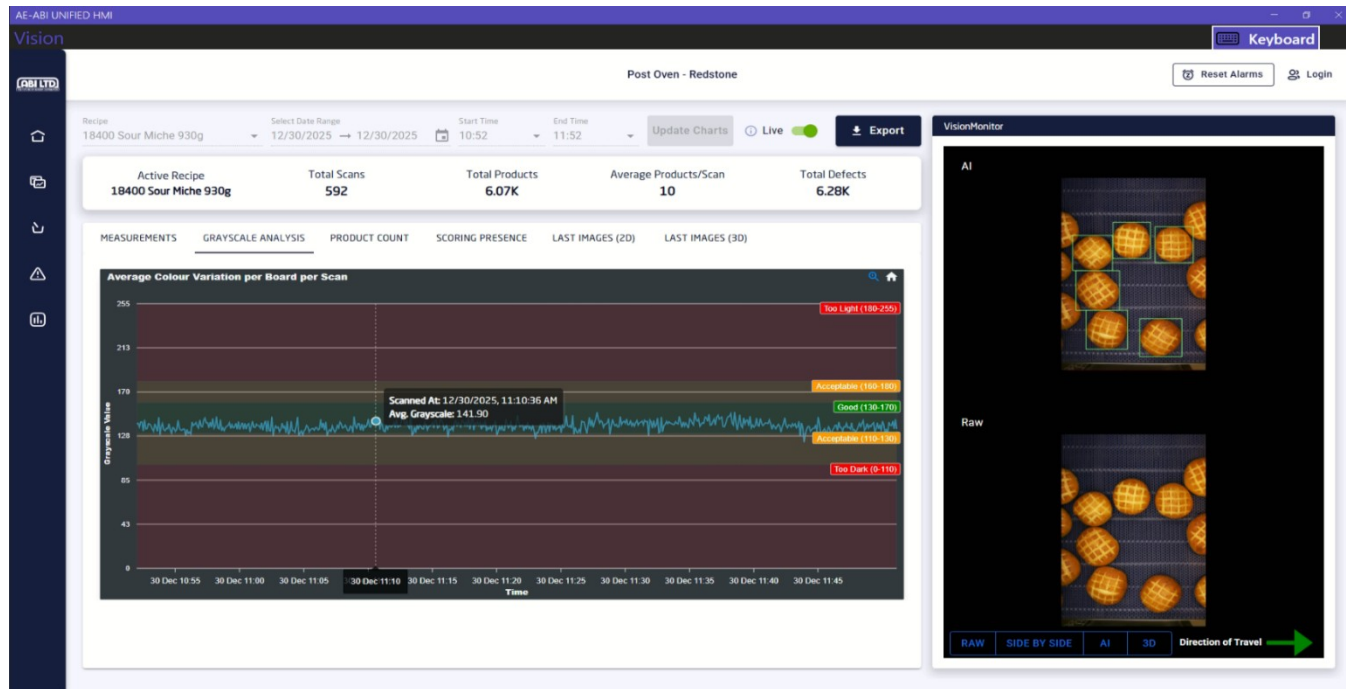
PRODUCT COUNT

LAST IMAGES (2D)

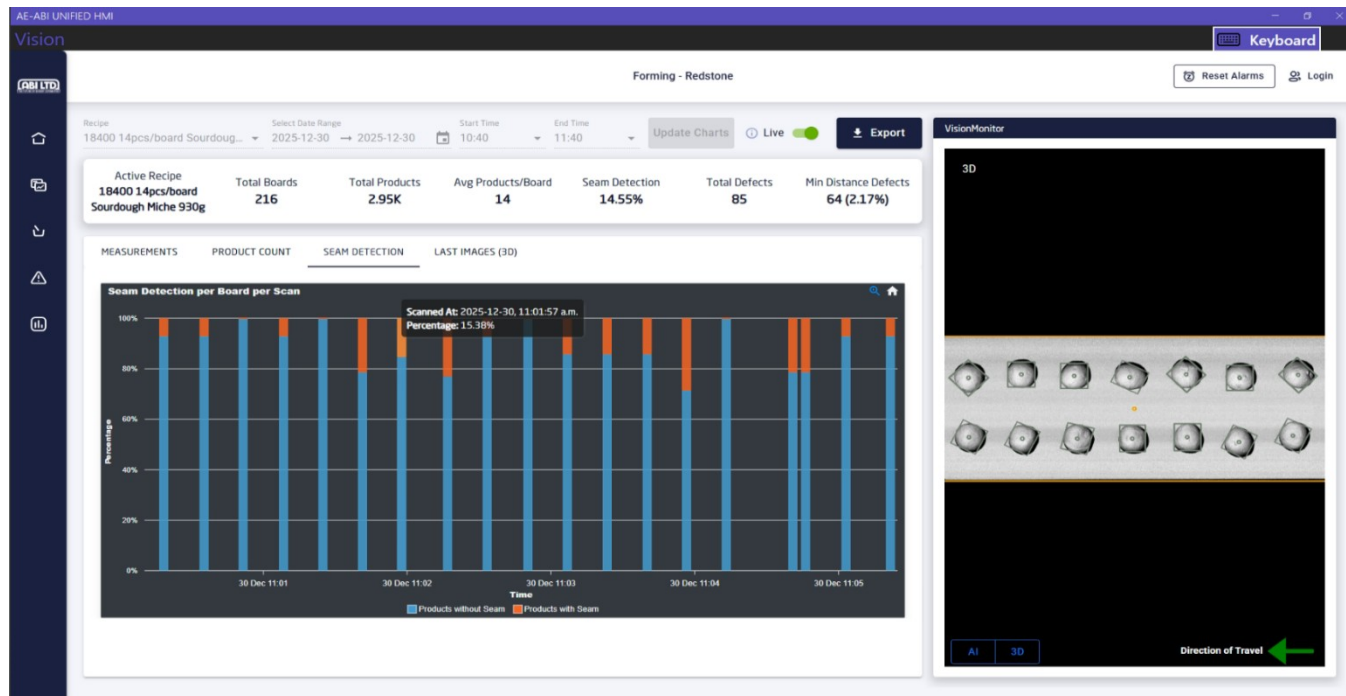
L Channel Variation Per Scan



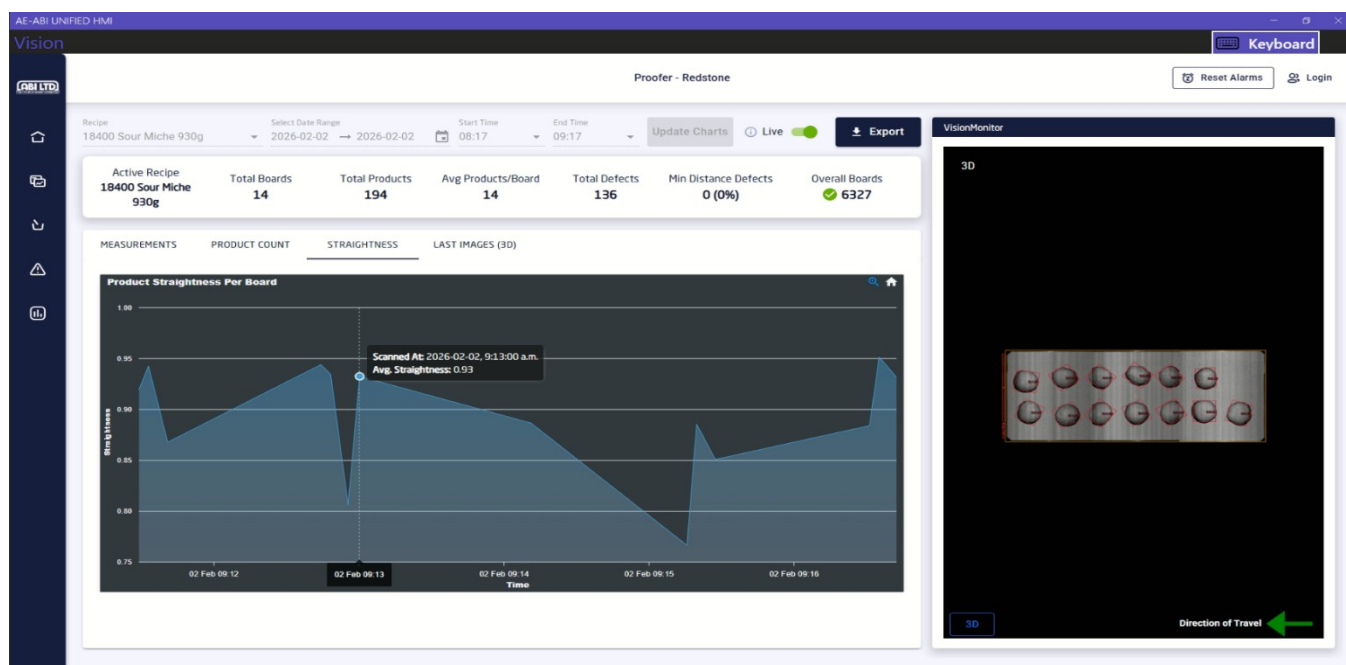
- **Grayscale Analysis (if configured):** Displays the average grayscale values (from 0 to 255) for products over time. The chart may include a legend to indicate ranges for good, acceptable, and not acceptable values, which are specific to each recipe and configured on the Thresholds page.



- **Seam Detection (if configured):** If your system is set up for it, this tab shows the percentage of products with detected seams versus those without seams, calculated per scan or board.



- **Straightness (if configured):** This tab displays the average ratio of product length to product width for products per scan or board.



- **Last Images (2D & 3D):** These tabs display images captured by the system within the last 24 hours. The most recent images are shown first, with options to navigate through older images. Clicking on an image will enlarge it for a clearer view.

Vision



Recipe

Default Recipe

Select Date Range

2026-04-27 → 2026-04-27

Start Time



09:16

Active Recipe
Default Recipe

Total Scans
295

Total Products
3.59K

Avg Products/Scan
12

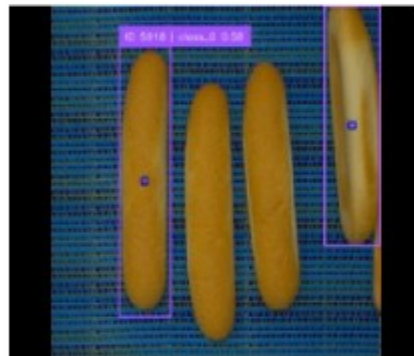
MEASUREMENTS

L CHANNEL ANALYSIS

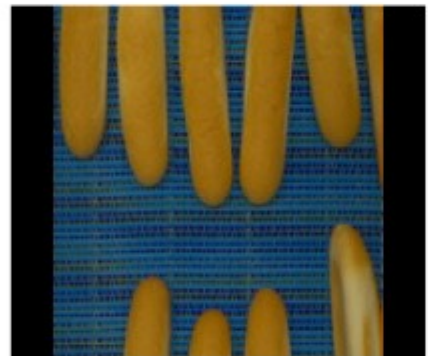
PRODUCT COUNT

LAST IMAGES (2D)

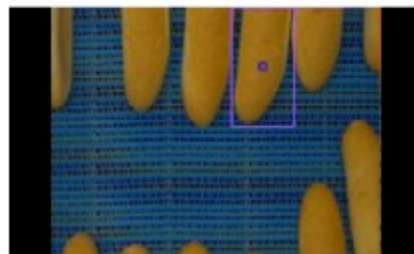
2026-04-27 10:15:20



2026-04-27 10:15:19

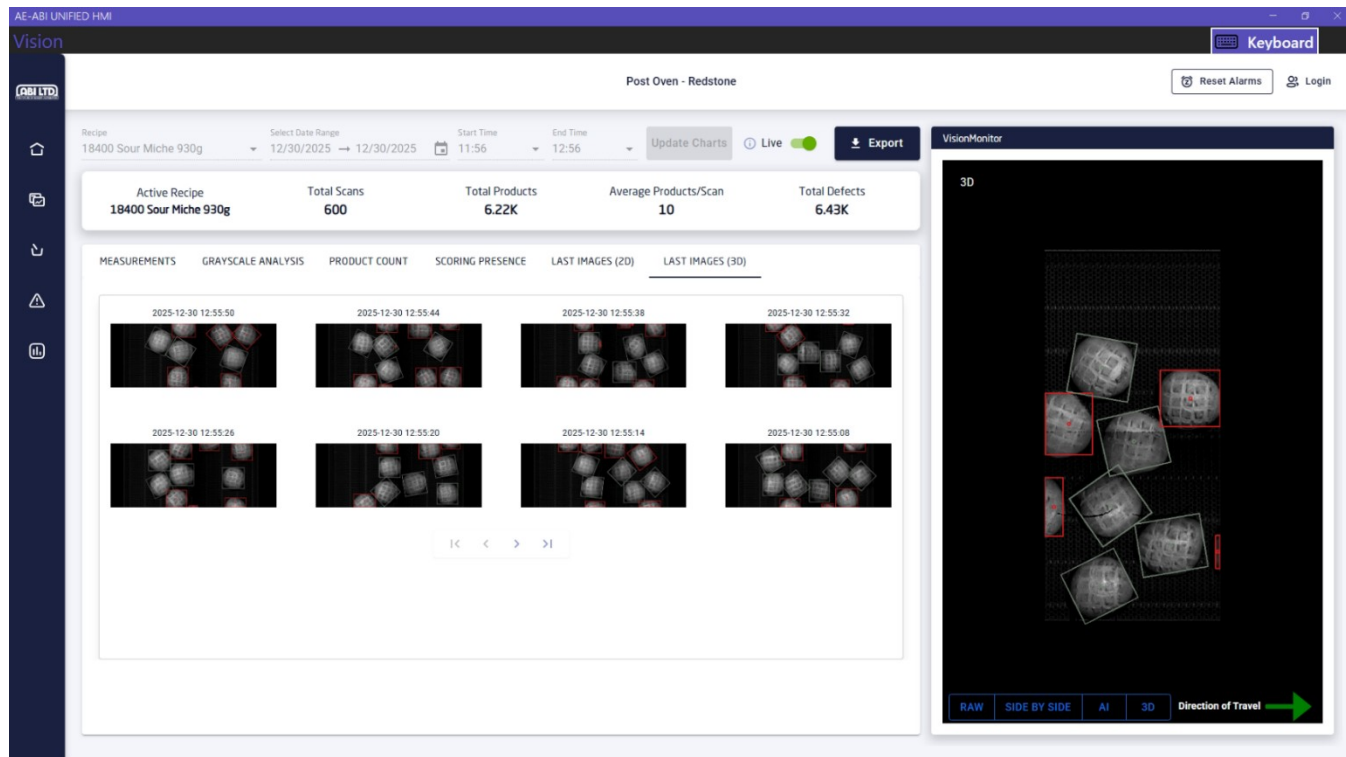


2026-04-27 10:15:16



2026-04-27 10:15:15

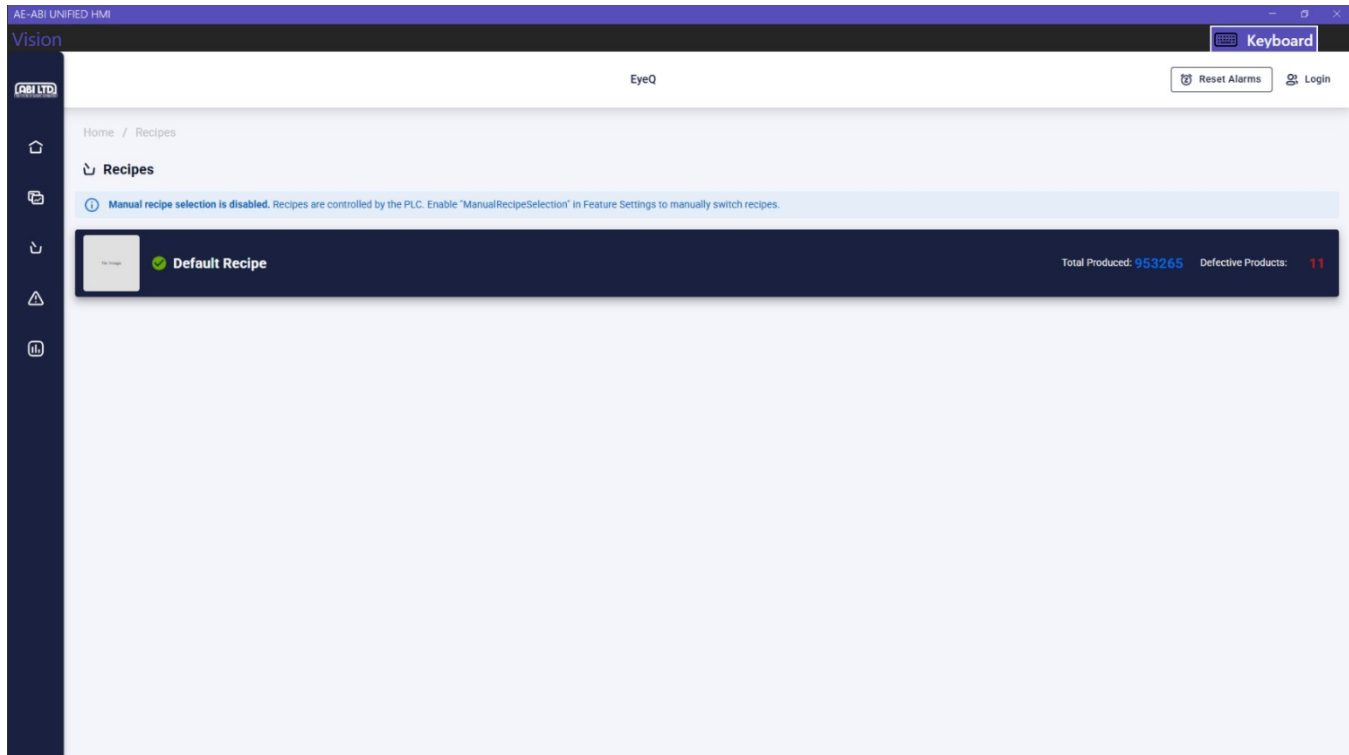




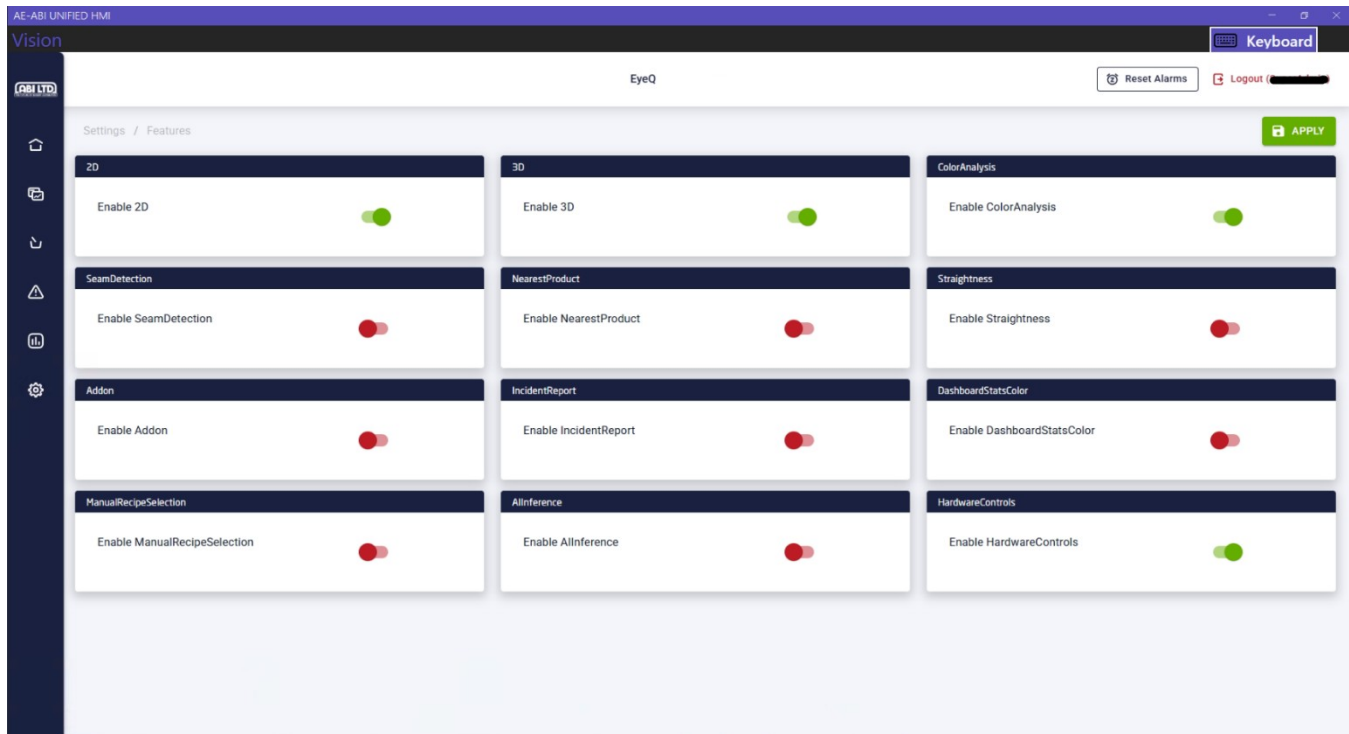
Recipes

The Recipes page lists all product recipes configured in your BDF system, along with their production data.

- **Recipe Overview:** For each recipe, you can see the total number of products produced and the number of products that had defects since that recipe was last active.
- **Active Recipe:** The recipe currently running on the system is clearly highlighted (e.g., with a darker background).



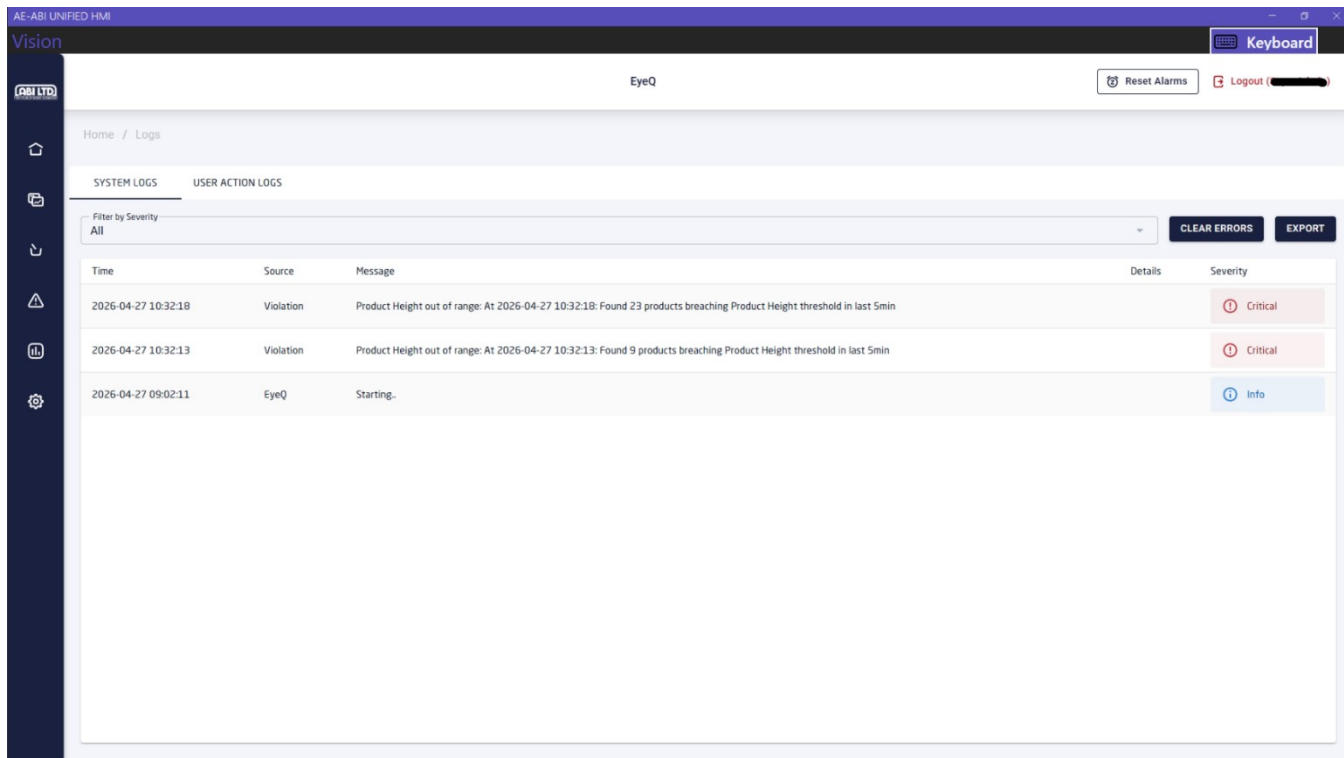
- **Manual Recipe Selection:** By default, the active recipe is controlled automatically by the system's automation (PLC). To enable manual recipe selection, you will need to log in as an administrator and activate the "ManualRecipeSelection" feature under 'Settings > Feature Settings'.



Logs

The Logs page provides important system records and user activity information.

- **System Logs:** Displays system events and details about product defects detected during image processing. These logs can be filtered by severity (e.g., informational, warning, error) and exported for review. A "Clear Errors" button is available on the right side of the page to clear current error messages.



The screenshot displays the 'Vision' software interface, specifically the 'EyeQ' logs page. The top navigation bar includes 'Home / Logs' and a 'Filter by Severity' dropdown menu set to 'All'. On the right, there are buttons for 'Reset Alarms', 'Logout', 'CLEAR ERRORS', and 'EXPORT'. The main content area is a table with the following data:

Time	Source	Message	Details	Severity
2026-04-27 10:32:18	Violation	Product Height out of range: At 2026-04-27 10:32:18: Found 23 products breaching Product Height threshold in last 5min		Critical
2026-04-27 10:32:13	Violation	Product Height out of range: At 2026-04-27 10:32:13: Found 9 products breaching Product Height threshold in last 5min		Critical
2026-04-27 09:02:11	EyeQ	Starting..		Info

- **User Action Logs:** This section tracks actions taken by users within the system, such as login and logout times, and any changes made to recipe thresholds. These logs can be filtered by start and end date, specific user, and type of action, and can also be exported.

Time	User	Action	Description	Recipe	Parameter	Old Value	New Value
2026-04-27 10:33:23	SuperAdmin	Logout	User 'SuperAdmin' logged out				
2026-04-27 10:32:42	SuperAdmin	ThresholdChange	Threshold 'Product Height' updated: IsActive: True → False, NotificationChannels: None → Banner, StackLight, Alarm	Default Recipe	Product Height	Min:80, Max:110, Target:100, Active:True	Min:80, Max:110, Target:100, Active:False
2026-04-27 10:31:59	SuperAdmin	ThresholdChange	Threshold 'Product Height' updated: IsActive: False → True, NotificationChannels: Banner, StackLight, Alarm → None	Default Recipe	Product Height	Min:80, Max:110, Target:100, Active:False	Min:80, Max:110, Target:100, Active:True
2026-04-27 10:31:16	SuperAdmin	Login	User 'superadmin' logged in				
2026-04-27 09:07:02	SuperAdmin	Logout	User 'SuperAdmin' logged out				
2026-04-27 09:02:49	SuperAdmin	Login	User 'superadmin' logged in				
2026-04-24 10:21:41	SuperAdmin	Logout	User 'SuperAdmin' logged out				
2026-04-24 10:14:45	SuperAdmin	Login	User 'superadmin' logged in				
2026-04-24 10:09:46	SuperAdmin	Logout	User 'SuperAdmin' logged out				
2026-04-24 09:43:14	SuperAdmin	Login	User 'superadmin' logged in				
2026-04-23 12:03:35	SuperAdmin	Logout	User 'SuperAdmin' logged out				
2026-04-23 10:22:17	SuperAdmin	ThresholdChange	Threshold 'Product Height' updated: IsActive: True → False	Default Recipe	Product Height	Min:80, Max:110, Target:100, Active:True	Min:80, Max:110, Target:100, Active:False
2026-04-23 10:21:47	SuperAdmin	ThresholdChange	Threshold 'Product Height' updated: IsActive: False → True	Default Recipe	Product Height	Min:80, Max:110, Target:100, Active:False	Min:80, Max:110, Target:100, Active:True

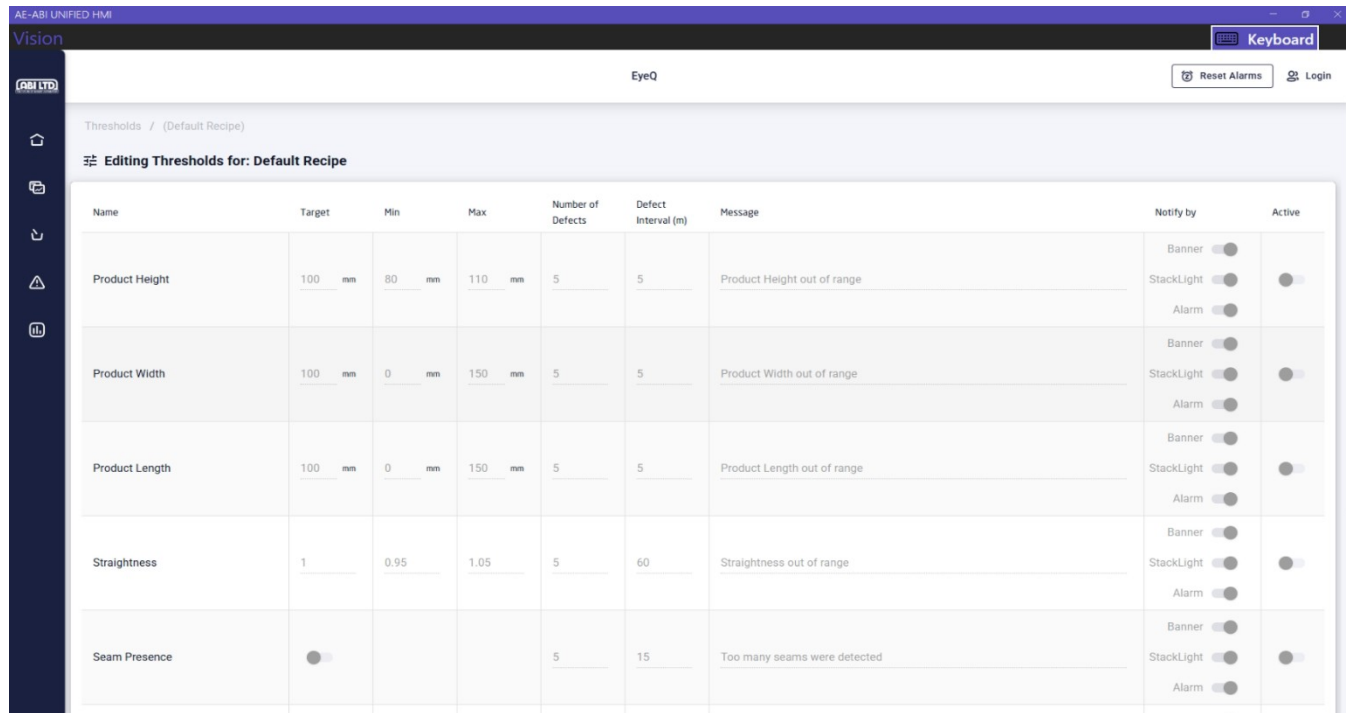
Thresholds

The Thresholds page allows you to view and configure various quality control parameters for the active recipe. The specific thresholds available may vary depending on how your BDF EyeQ system is configured for your bakery products.

Each threshold includes the following important fields:

- **Name:** The label for the threshold (e.g., Product Height, Product Width, Product Length, Straightness, Seam Presence, Minimum Distance Between Products, Products Per Scan/Board, Scans/Boards Per Hour, Product Grayscale).
- **Target:** The ideal value you want your products to achieve for this parameter.
- **Min:** The lowest acceptable value for the threshold.
- **Max:** The highest acceptable value for the threshold.
- **Number of Defects:** The count of defects that must occur before the system raises an alarm.
- **Defect Interval (min):** The duration (in minutes) over which the system monitors for the "Number of Defects." For example, if "Number of Defects" is 50 and "Defect Interval" is 5 minutes, an alarm will be raised if 50 or more defects for that threshold are detected within any 5-minute period.
- **Message:** The text message that will be displayed when an alarm is triggered.
- **Notify by:** The channels through which you will be alerted (e.g., a banner on screen, a stack light, an alarm buzzer). A "Reset Alarms" button is located at the top right to clear all active alarms.

- **Active:** Indicates whether this specific threshold is currently enabled for monitoring and logging defects in the BDF EyeQ application.



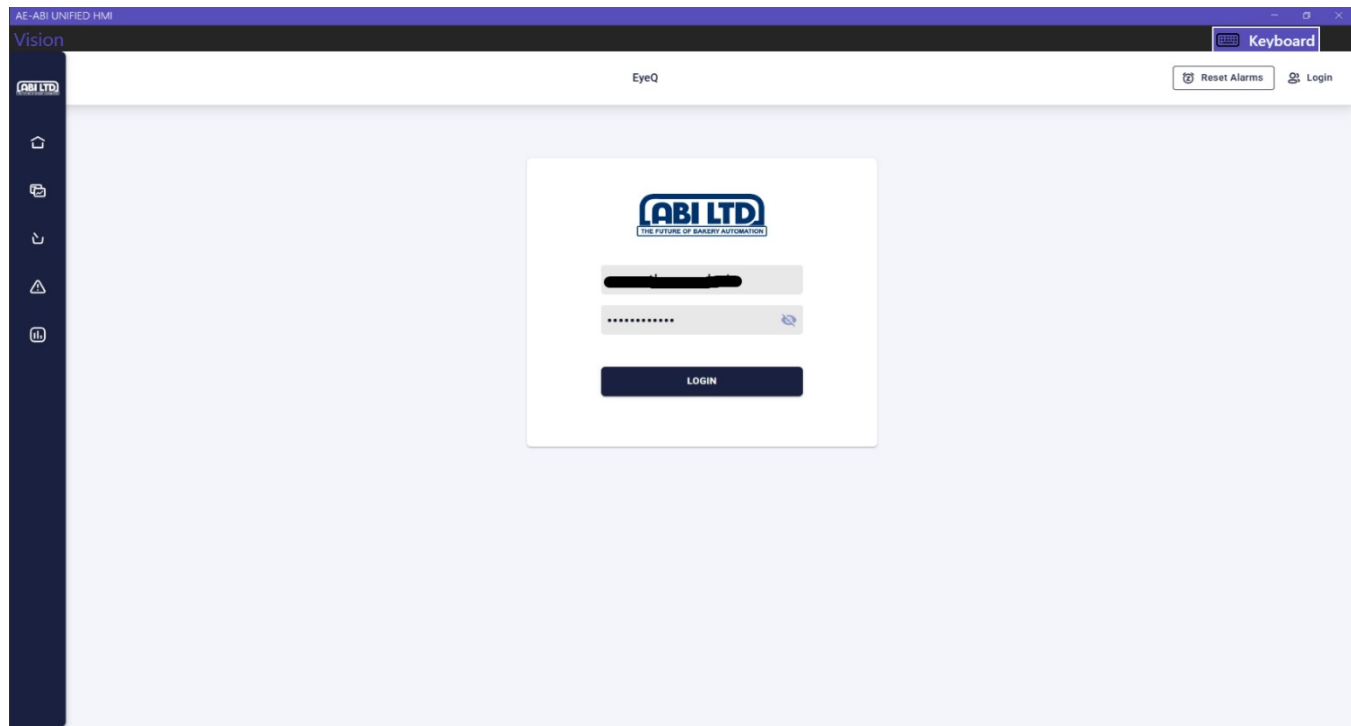
Name	Target	Min	Max	Number of Defects	Defect Interval (m)	Message	Notify by	Active
Product Height	100 mm	80 mm	110 mm	5	5	Product Height out of range	Banner ☐ StackLight ☐ Alarm ☐	☐
Product Width	100 mm	0 mm	150 mm	5	5	Product Width out of range	Banner ☐ StackLight ☐ Alarm ☐	☐
Product Length	100 mm	0 mm	150 mm	5	5	Product Length out of range	Banner ☐ StackLight ☐ Alarm ☐	☐
Straightness	1	0.95	1.05	5	60	Straightness out of range	Banner ☐ StackLight ☐ Alarm ☐	☐
Seam Presence	☐			5	15	Too many seams were detected	Banner ☐ StackLight ☐ Alarm ☐	☐

By default, regular users can only view the threshold settings. To make changes, you must log in with administrator privileges.

Login

The Login button is conveniently located at the top-right corner of the BDF EyeQ application. Logging in as an administrator is essential to make changes to critical system settings, such as modifying threshold values, configuring camera settings, or adjusting general system parameters.

Your specific administrator login credentials will be provided by your BDF system administrator.

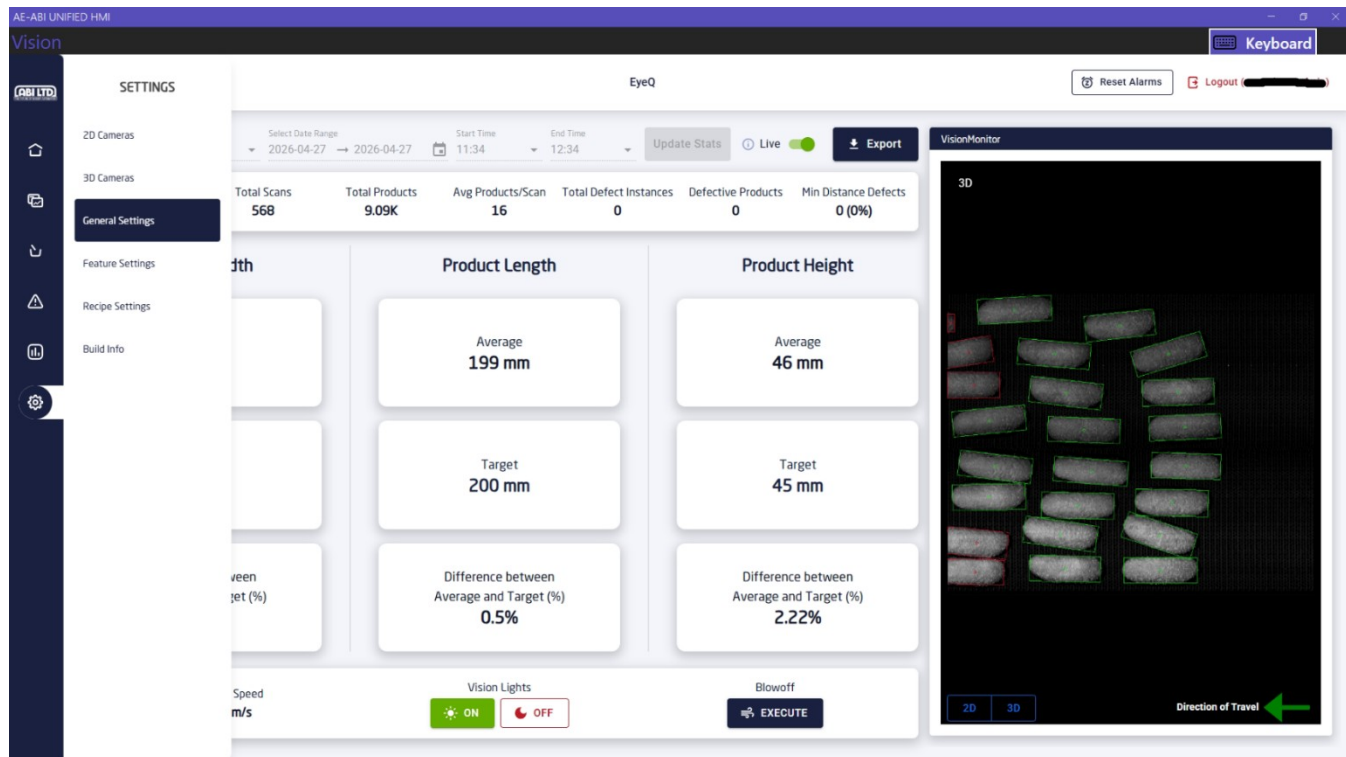


Settings (Requires Administrator Login)

The following settings pages are accessible only after you have successfully logged in with administrator privileges.

General Settings

Access this page by navigating to 'Settings > General Settings'. This section allows you to configure fundamental system parameters for your BDF EyeQ application.



- **Company and Machine Details:** Here, you can set the company name (BDF), the specific machine name, and specify the storage location for images captured by the application.

The screenshot shows the 'Settings / General' page. It has a sidebar with navigation icons and an 'APPLY' button. The settings are organized into sections:

- Machine Details**
 - Company Name: [Empty field]
 - Machine Name: EyeQ
- Paths Configuration**
 - Configs Folder Path: C:\ProgramData\Aetherics\Unity
 - 2D Images Path: C:\ProgramData\Aetherics\Unity\2D_Data
 - 3D Images Path: C:\ProgramData\Aetherics\Unity\3D_Data
- Capture Mode**
 - Time delay (ms): 1000
 - Capture Mode Enable: [Red toggle switch] Disabled

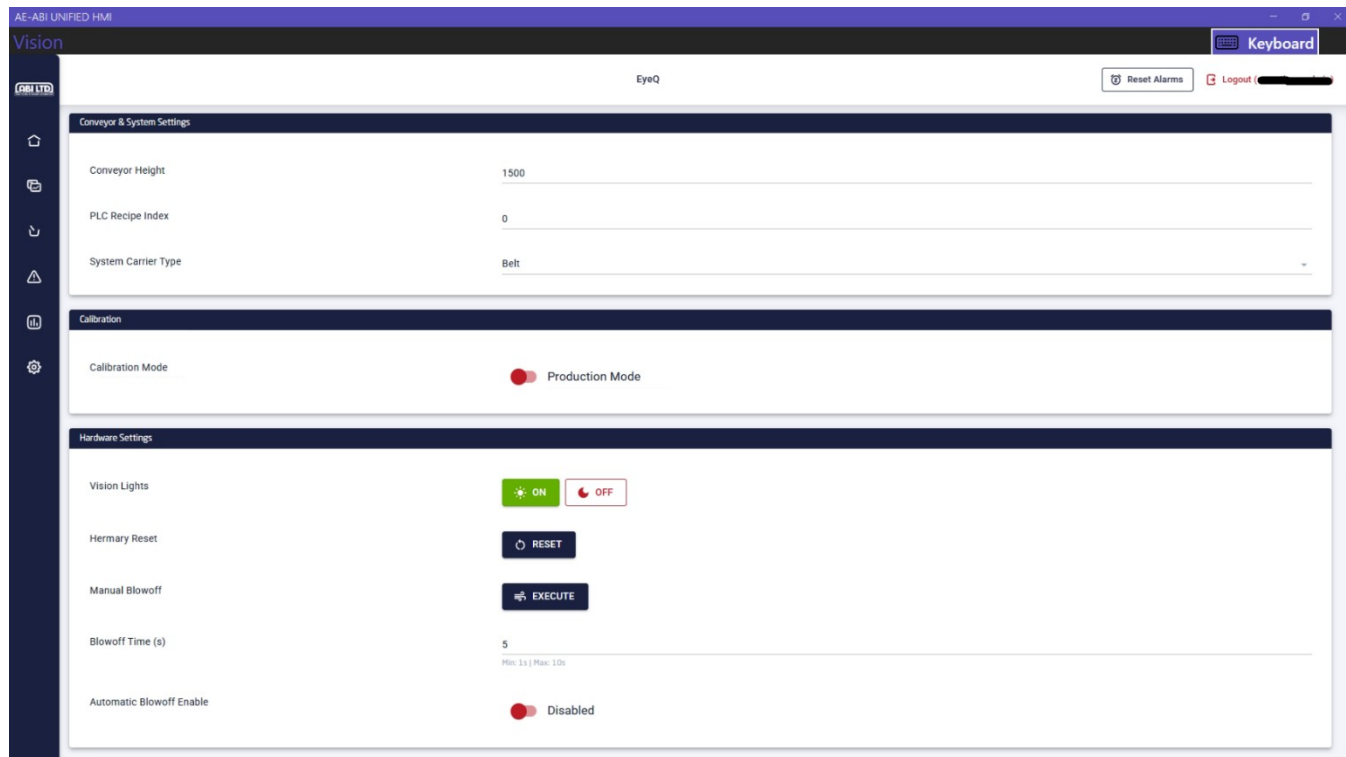
- **Image Capture Settings:** Configure how images are previewed, select whether image capture is triggered by software or hardware, and define the scan line source (either time-based or encoder-based).

The screenshot displays the 'Vision' configuration window for the EyeQ system. The interface is divided into three main sections:

- Preview Settings:**
 - Preview Rotation Angle (CW): 270
 - Travel Direction Angle (CW): 270
 - Preview Scale (0.8x): A slider control.
- Trigger Settings:**
 - Trigger Enable: ☒ Enabled
 - Trigger Delay (ms): 4000
 - Trigger Source: DAQ_Time (dropdown menu)
 - Software Trigger Enable: ☒ Enabled
- Scan Line Source:**
 - Scan Line Source: ☒ Time-Based

The top of the window shows 'AE-ABI UNIFIED HMI' and 'Vision'. The right side has a 'Keyboard' button and a 'Reset Alarms' button. The bottom right corner shows a 'Logout' button.

- **Conveyor and Hardware Settings:** This includes settings for conveyor height (in mils), the type of product carrier (belt or pan), and the ability to switch between production and calibration modes. It also provides controls for certain hardware components like turning vision lights on/off, resetting cameras, and blowoff settings. Please note that while these values can be adjusted, the current settings are optimized to ensure the highest performance of your BDF EyeQ system.



Feature Settings

Access this page by navigating to `Settings > Feature Settings`. This section allows you to enable or disable various operational features within the BDF EyeQ application.

Vision



Settings / Features

2D

Enable 2D



SeamDetection

Enable SeamDetection



Addon

Enable Addon



ManualRecipeSelection

Enable ManualRecipeSelection



3D

Enable 3D

NearestF

Enable NearestF

IncidentF

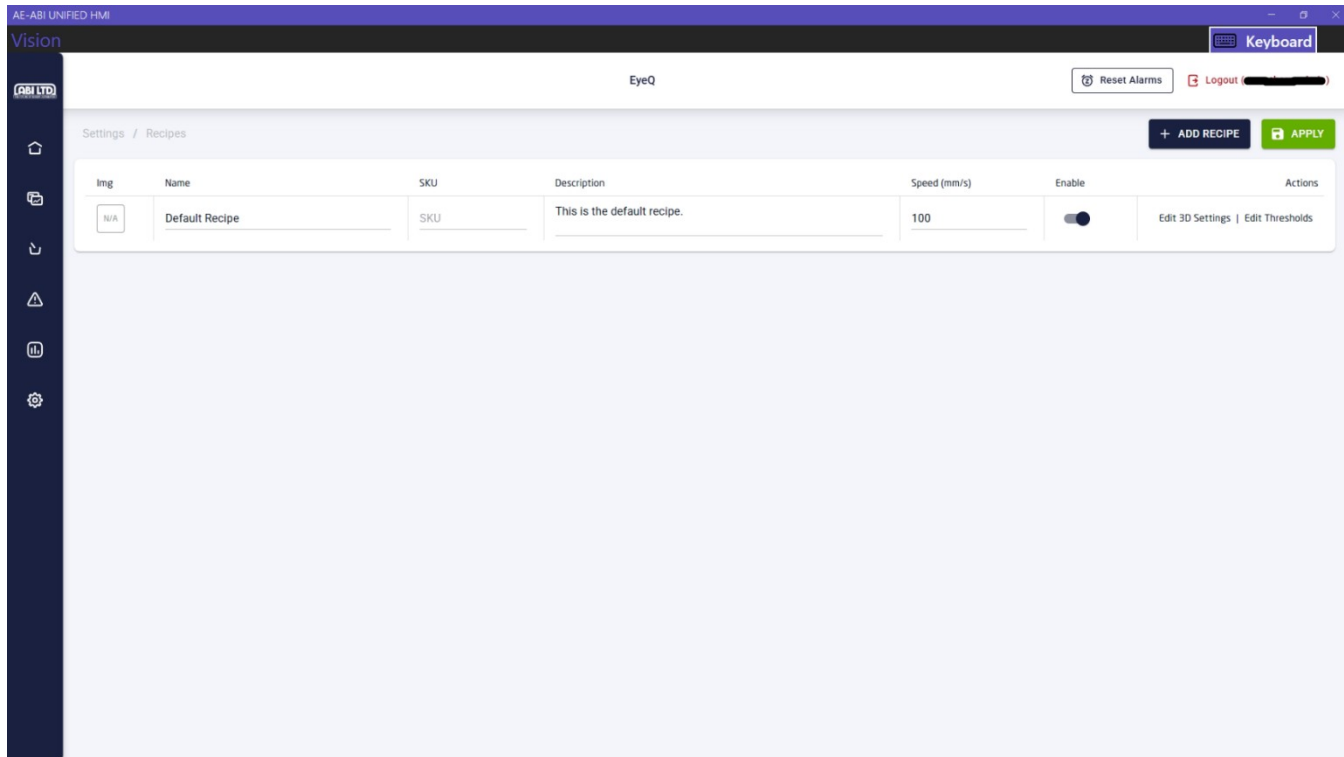
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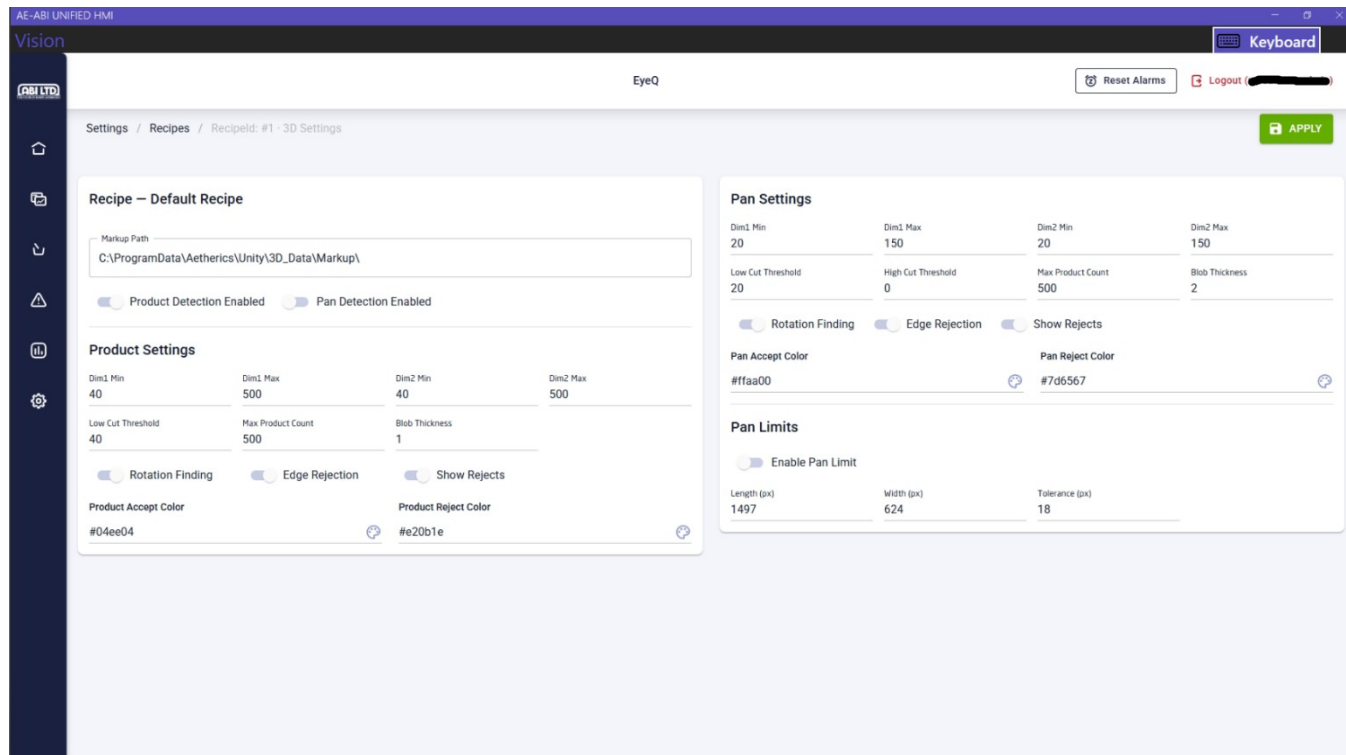
Enable AllInferen

Recipe Settings

Access this page by navigating to 'Settings > Recipe Settings'. This section lists all configured recipes and enables you to modify their specific parameters.



- **Edit 3D Settings:** Click the "Edit 3D settings" button for a particular recipe to adjust specific capture settings. This includes defining minimum and maximum acceptable product dimensions, setting the colors used for product acceptance and rejection, and configuring pan settings (if your bakery products are conveyed on pans). Remember to click the "APPLY" button to save any changes you make.



- Edit Thresholds:** Click the "Edit Thresholds" button for a specific recipe to set or modify the quality control values. After making your desired changes, click the "APPLY" button to save them.



Thresholds / (Default Recipe)

Editing Thresholds for: Default Recipe

Name	Target	Min	Max	
Product Height	100mm	80mm	110mm	
Product Width	100mm	0mm	150mm	
Product Length	100mm	0mm	150mm	
Straightness	1	0.95	1.05	